

PROGRAM 1: IPC USING PIPE

C PROGRAM

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <unistd.h>
```

```
#include <string.h>
```

```
#include <sys/wait.h>
```

```
int main()
```

```
{
```

```
    int fd[2];
```

```
    pid_t pid;
```

```
    char message[] = "Hello from Child Process";
```

```
    char buffer[100];
```

```
    pipe(fd);
```

```
    pid = fork();
```

```
    if (pid == 0)
```

```
    {
```

```
        close(fd[0]);
```

```
        write(fd[1], message, strlen(message) + 1);
```

```
        close(fd[1]);
```

```
        exit(0);
```

```
    }
```

```
    else {
```

```
        wait(NULL);
```

```
        close(fd[1]);
```

```
        read(fd[0], buffer, sizeof(buffer));
```

```
        printf("Message received from child: %s\n", buffer);
```

```
        close(fd[0]);    }
```

```
    return 0;
```

}
SHELL SCRIPT

```
GNU nano 8.7.1
#!/bin/bash

(
    echo "Message from Child Process"
) | while read msg
do
    echo "Parent Received: $msg"
done
```

Output :

```
(ganesh@LAPTOP-R3RDH9NG)~$ nano ipc.sh

(ganesh@LAPTOP-R3RDH9NG)~$ chmod +x ipc.sh

(ganesh@LAPTOP-R3RDH9NG)~$ bash ipc.sh
Parent Received: Message from Child Process
```